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### Liquid ejection head

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#### Abstract

A liquid ejection head includes a plurality of nozzles arranged along a first direction of an ejection surface, a plurality of pressure chambers communicating with the plurality of individual nozzles and provided with actuators configured to apply, to a liquid, pressures for ejecting the liquid from the nozzles, a plurality of discrete flow paths communicating with the plurality of individual pressure chambers, a common flow path communicating with the plurality of individual discrete flow paths via discrete opening portions and extending in the first direction, and a damper mechanism disposed to face the common flow path in a direction crossing the ejection surface to absorb a pressure fluctuation in the liquid in the common flow path. The common flow path is provided with a partition wall disposed between the discrete opening portions adjacent in the first direction to extend in the direction crossing the ejection surface.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC          |
|--------------|-------------|---------------|----------|--------------|
| 8662650      | 12/2013     | Ono et al.    | N/A      | N/A          |
| 11123985     | 12/2020     | Mizuno        | N/A      | B41J 2/14233 |
| 2019/0232659 | 12/2018     | Takabe        | N/A      | H10N 30/883  |
| 2022/0227133 | 12/2021     | Knechten      | N/A      | H10N 30/875  |

### FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 2006-347036 | 12/2005          | JP      | N/A |
| 2013-203062 | 12/2012          | JP      | N/A |

### OTHER PUBLICATIONS

U.S. Appl. No. 18/126,664, Akiko Hammura Yoshiyuki Nakagawa Takura Yamazaki Atsushi Teranishi, filed Mar. 27, 2023. cited by applicant

U.S. Appl. No. 18/126,687, Yoshiyuki Nakagawa Takura Yamazaki Akiko Hammura Atsushi Teranishi, filed Mar. 27, 2023. cited by applicant

U.S. Appl. No. 18/126,674, Akiko Hammura Yoshiyuki Nakagawa Takuro Yamazaki Atsushi Teranishi, filed Mar. 27, 2023. cited by applicant

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## Background/Summary

### BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a liquid ejection head.

Description of the Related Art

(2) In a liquid ejection head to be used in a liquid ejection apparatus that ejects a liquid such as ink, an ejection surface is provided with a plurality of nozzles, and a pressure chamber having an actuator such as a piezoelectric element that applies a pressure to the liquid to eject droplets from the nozzles is provided. A pressure fluctuation applied to the liquid in the pressure chamber for droplet ejection may affect characteristics of subsequent droplet ejection. The droplet ejection characteristics include an ejection speed, a droplet volume, a state of coherence of droplets, and the like. When the droplet ejection characteristics change, droplet landing positions on a paper surface, an area of the droplets after landing, the number of the landing droplets, and the like change in a printing apparatus using the liquid ejection head to affect a printing quality. Meanwhile, a technology including a damper mechanism that absorbs a pressure fluctuation in a liquid resulting from droplet ejection is known. Each of Japanese Patent Application Publication No. 2006-347036 and Japanese Patent Application Publication No. 2013-203062 describes a technology of disposing a damper chamber at a position facing a flow path for a liquid in a direction crossing an ejection surface, disposing a flexible damper wall between the damper chamber and the flow path, and using warping of the damper wall to absorb a pressure fluctuation in the liquid in the flow path.

#### SUMMARY OF THE INVENTION

(3) A configuration of a liquid ejection head is known that includes a common flow path commonly communicating with a plurality of pressure chambers individually communicating with a plurality of nozzles and extending in a direction in which the plurality of nozzles are arranged, supplies a liquid from the outside to the common flow path, and distributes the liquid from the common flow path to the individual pressure chambers. In the technology of each of Japanese Patent Application Publication No. 2006-347036 and Japanese Patent Application Publication No. 2013-203062, a damper mechanism facing such a common flow path in the direction crossing the ejection surface is disposed. However, such a damper mechanism cannot sufficiently inhibit propagation of the pressure fluctuation in the direction in which the plurality of nozzles are arranged in the common flow path. Due to the pressure fluctuation propagating in the direction in which the plurality of nozzles are arranged in the common flow path, a pressure fluctuation occurring in a given one of the pressure chambers may affect characteristics of droplet ejection from the nozzles of another of the pressure chambers communicating therewith via the common flow path.

(4) In view of the foregoing problem to be solved, an object of the present invention is to inhibit, in a liquid ejection head in which a plurality of nozzles communicate with each other via a common flow path, a pressure fluctuation resulting from droplet ejection from affecting droplet ejection from another nozzle communicating via the common flow path.

(5) The present invention is a liquid ejection head configured to eject a liquid from an ejection surface, the liquid ejection head comprising: a plurality of nozzles arranged along a first direction of the ejection surface; a plurality of pressure chambers communicating with the plurality of individual nozzles and provided with actuators configured to apply, to the liquid, pressures for ejecting the liquid from the nozzles; a plurality of discrete flow paths communicating with the plurality of individual pressure chambers; a common flow path communicating with the plurality of individual discrete flow paths via discrete opening portions and extending in the first direction; and a damper mechanism disposed to face the common flow path in a direction crossing the ejection surface to absorb a pressure fluctuation in the liquid in the common flow path, the common flow path being provided with a partition wall disposed between the discrete opening portions adjacent in the first direction to extend in the direction crossing the ejection surface.

(6) The present invention is a liquid ejection head configured to eject a liquid from an ejection surface, the liquid ejection head comprising: a plurality of nozzles arranged along a first direction of the ejection surface; a plurality of pressure chambers communicating with the plurality of individual nozzles and provided with actuators configured to apply, to the liquid, pressures for ejecting the liquid from the nozzles; a plurality of discrete flow paths communicating with the plurality of individual pressure chambers; a common flow path communicating with the plurality of

individual discrete flow paths and extending in the first direction; and a damper mechanism disposed to face the common flow path in a direction crossing the ejection surface to absorb a pressure fluctuation in the liquid in the common flow path, the common flow path being provided with a partition wall that partially blocks a flow of the liquid along the first direction.

(7) According to the present invention, it is possible to inhibit, in the liquid ejection head in which the plurality of nozzles communicate with each other via the common flow path, the pressure fluctuation resulting from droplet ejection from affecting droplet ejection from another nozzle communicating via the common flow path.

(8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram of an inkjet recording apparatus;

(2) FIG. 2A is a schematic view of a liquid ejection head module;

(3) FIGS. 2B and 2C are schematic views of the liquid ejection head module;

(4) FIG. 3A is a schematic cross-sectional view of a liquid ejection substrate;

(5) FIG. 3B is a schematic cross-sectional view of the liquid ejection substrate;

(6) FIG. 4 is a schematic perspective plan view of the liquid ejection substrate;

(7) FIG. 5A is a schematic diagram of the liquid ejection substrate according to a first embodiment of the present invention;

(8) FIG. 5B is a schematic diagram of the liquid ejection substrate according to the first embodiment of the present invention;

(9) FIG. 6A is a schematic diagram of the liquid ejection substrate according to a second embodiment of the present invention;

(10) FIG. 6B is a schematic diagram of the liquid ejection substrate according to the second embodiment of the present invention;

(11) FIG. 7A is a schematic diagram of the liquid ejection substrate according to a third embodiment of the present invention;

(12) FIG. 7B is a schematic diagram of the liquid ejection substrate according to the third embodiment of the present invention;

(13) FIG. 8A is a schematic diagram of the liquid ejection substrate according to a fourth embodiment of the present invention; and

(14) FIG. 8B is a schematic diagram of the liquid ejection substrate according to the fourth embodiment of the present invention.

### DESCRIPTION OF THE EMBODIMENTS

(15) Referring to the drawings, a description will be given below of a liquid ejection head and a liquid ejection apparatus according to each of embodiments of the present invention. The following will describe the embodiment in which the present invention is applied to an inkjet recording head that ejects ink as an example of a liquid and to an inkjet recording apparatus, but the present invention is also applicable to another apparatus. For example, the present invention is applicable to apparatuses such as a printer, a copier, a fax machine having a communication system, and a word processor having a printer unit and to industrial recording apparatuses compositely combined with various processing apparatuses, e.g., apparatus that perform biochip fabrication, electronic circuit printing, and the like. A configuration in each of the following embodiments is for illustrative purposes, and various combinations and modifications are possible within the scope of the present invention.

(16) Description of Entire Head

(17) FIG. 1 is a schematic configuration diagram of an inkjet recording apparatus **101** according to each of the embodiments. The inkjet recording apparatus **101** is a one-pass type recording apparatus that uses a liquid ejection head module **1** to record an image on a recording medium **111** while the recording medium **111** is transported once by a transport unit **110**. It is assumed hereinbelow that a width direction of the recording medium **111** is an X-direction, a direction (indicated by an arrow A) in which the recording medium **111** is transported is a Y-direction, and a direction crossing each of the X-direction and the Y-direction is a Z-direction. The X-direction and the Y-direction are directions along an ejection surface formed with nozzles of the liquid ejection head module **1** (described later), the X-direction (first direction) is a direction in which the nozzles are arranged, and the Y-direction (second direction) is a direction in which nozzle rows are arranged. The Y-direction (second direction) is a direction crossing the X-direction (first direction) along the ejection surface. Typically, the X-direction and the Y-direction are perpendicular to each other along a horizontal plane, while the Z-axis is parallel to a vertical direction.

(18) The liquid ejection head module **1** includes discrete modules that eject cyan, magenta, yellow, and black inks. In the case of distinguishing the liquid ejection head modules for the individual colors from each other, the liquid ejection head modules are provided with marks C, M, Y, and K to be distinguished from each other. The liquid ejection head modules for the four colors are arranged along the direction (Y-direction) in which the recording medium **111** is transported. Each of the liquid ejection head modules for the individual colors has sub-modules arranged along the width direction (X-direction) of the recording medium **111**. In the case of distinguishing the sub-modules from each other, the sub-modules are provided with marks a and b to be distinguished from each other. In FIG. 1, the liquid ejection head module **1** is disposed vertically above the recording medium **111** to eject the inks vertically downward (in the Z-direction). Note that a configuration of the liquid ejection head module **1** illustrated in FIG. 1 is exemplary, and the present invention is also applicable to a liquid ejection head module in another form.

(19) Description of Configuration of Liquid Ejection Head

(20) FIGS. 2A to 2C are schematic views of the liquid ejection head module **1**. FIG. 2A is a perspective view obtained by viewing the liquid ejection head module **1** from an ejection surface side. FIG. 2B is a diagram illustrating the ejection surface of each of liquid ejection substrates **2**. FIG. 2C is a diagram illustrating a surface of the liquid ejection substrate **2** opposite to the ejection surface.

(21) The liquid ejection head module **1** has a head main body **4** and the plurality of liquid ejection substrates **2** disposed in the head main body **4**. The liquid ejection head module **1** has a plurality of nozzles **3** arranged in the X-direction (first direction) along ejection surfaces **30** of the liquid ejection substrates **2**.

(22) Each of the liquid ejection substrates **2** has a nozzle substrate **201**, and the plurality of nozzles **3** are arranged along a longitudinal direction (X-direction) of the nozzle substrate **201** to form the nozzle rows. In the nozzle substrate **201**, the plurality of nozzle rows are arranged along a transverse direction (Y-direction). Each of the liquid ejection substrates **2** has a flow path formation substrate **204** and, via external supply opening portions **20** formed in the flow path formation substrate **204**, the ink is supplied from an external ink tank to the liquid ejection substrate **2**. The supplied ink flows in flow paths in the liquid ejection substrate **2** to be ejected from the nozzles **3** and drop onto the recording medium **111**. To the plurality of external supply opening portions **20**, the ink is supplied from the ink tank (not shown) via a common feeding port (not shown) provided in the head main body **4**.

(23) In the head main body **4**, an electric circuit substrate (not shown) for supplying electric power and a signal for driving actuators such as piezoelectric elements that eject the ink from the nozzles **3** is disposed. The electric circuit substrate is connected via wiring (not shown) to terminals **10** of a vibration substrate **202** provided with the actuators of the liquid ejection substrate **2**. Note that a configuration of the liquid ejection head module **1** illustrated in FIGS. 2A to 2C is exemplary, and

the present invention is also applicable to a liquid ejection head module in another form.

(24) Description of Configuration of Liquid Ejection Substrate

(25) FIGS. 3A and 3B are schematic cross-sectional views of each of the liquid ejection substrates 2. FIG. 3A illustrates a B-B cross section in FIG. 2B. FIG. 3B is a diagram illustrating a part of FIG. 3A in enlarged relation.

(26) The liquid ejection substrate 2 is configured by joining together four substrates, i.e., the nozzle substrate 201, the vibration substrate 202, a liquid supply substrate 203, and the flow path formation substrate 204. Between the liquid supply substrate 203 and the flow path formation substrate 204, a damper member 300 is interposed. The flow path formation substrate 204 is provided with depressed portions extending in the X-direction (first direction), and the damper member 300 and the depressed portions form damper chambers each corresponding to a space extending in the X-direction (first direction). This space serves as a space in which a gas (which is typically atmospheric air) is present. This space and the damper member 300 are included in each of damper mechanisms 301 that absorbs a pressure fluctuation in the liquid.

(27) A plurality of pressure chambers 11 are formed to communicate with each of the plurality of nozzles 3, and the plurality of pressure chambers 11 are individually provided with piezoelectric elements 18 serving as the actuators that apply, to the liquid, pressures for ejecting the liquid from the nozzles 3. The piezoelectric elements 18 are provided in deformable wall surfaces made of the vibration substrate 202, and any of the piezoelectric elements 18 deforms the vibration substrate 202 to thereby apply a pressure to the liquid in the pressure chamber 11 and eject droplets from the nozzles 3.

(28) A plurality of discrete flow paths are formed to individually communicate with the plurality of individual pressure chambers 11, and a common flow path is formed to communicate with the plurality of individual discrete flow paths via discrete opening portions and extend in the X-direction (first direction). In the example in FIG. 3B, the discrete flow paths include discrete supply flow paths 12a that supply the liquid to the pressure chambers 11 and discrete discharge flow paths 12b that discharge the liquid from the pressure chambers 11. The common flow path is formed in the liquid supply substrate 203. In the example in FIG. 3B, the common flow path includes common supply flow paths 13a individually communicating with the plurality of discrete supply flow paths 12a via discrete supply opening portions 120a and common discharge flow paths 13b individually communicating with the plurality of discrete discharge flow paths 12b via discrete discharge opening portions 120b.

(29) The damper member 300 is disposed to face the common flow path in the direction (Z-direction) crossing the ejection surface 30 to absorb a pressure fluctuation in the liquid in the common flow path. In the example in FIG. 3B, wall surfaces of the common supply flow paths 13a facing the discrete supply flow paths 12a are formed of the damper member 300, and accordingly the damper mechanisms 301 are disposed at positions facing the discrete supply flow paths 12a. Meanwhile, wall surfaces of the common discharge flow paths 13b facing the discrete discharge flow paths 12b are formed of the damper member 300, and accordingly the damper mechanisms 301 are disposed at positions facing the discrete discharge flow paths 12b. Note that the damper member 300 may also be configured to be disposed to face at least either of the common supply flow path 13a and the common discharge flow path 13b in the direction (Z-direction) crossing the ejection surface 30.

(30) The plurality of common supply flow paths 13a individually communicate with a plurality of supply connection flow paths 15a formed in the flow path formation substrate 204. The plurality of supply connection flow paths 15a are individually formed with a plurality of external supply opening portions 20a, and the liquid is supplied from the outside via the external supply opening portions 20a. The plurality of common discharge flow paths 13b individually communicate with a plurality of discharge connection flow paths 15b formed in the flow path formation substrate 204. The plurality of discharge connection flow paths 15b are individually formed with a plurality of

external discharge opening portions **20b**, and the liquid is discharged to the outside via the external discharge opening portions **20b**.

(31) Each of the nozzle substrate **201**, the vibration substrate **202**, the liquid supply substrate **203**, and the flow path formation substrate **204** is made of a silicon substrate or the like. Note that a configuration and the number of the substrates forming each of the liquid ejection substrates **2** are not limited to those in this example. The damper member **300** is formed of an elastic material, and a resin material such as, e.g., polyimide or polyamide can be used.

(32) FIG. **4** is a perspective plan view illustrating a part of the liquid ejection substrate **2**. The liquid ejection substrate **2** has the plurality of nozzles **3** and the plurality of pressure chambers **11** connected individually to the plurality of nozzles **3**. The plurality of nozzles **3** are arranged along the X-direction (first direction) to form one nozzle row, and a plurality of such nozzle rows are formed along the Y-direction (second direction). Both ends of each of the pressure chambers **11** in the longitudinal direction (Y-direction) are formed with the discrete supply opening portion **120a** communicating with the discrete supply flow path **12a** and with the discrete discharge opening portion **120b** communicating with the discrete discharge flow path **12b**. In positional relationships along the ejection surface (XY plane), each of the damper mechanisms **301** is disposed so as to include positions of the discrete supply opening portions **120a** and each of the damper mechanisms **301** is disposed so as to include positions of the discrete discharge opening portions **120b**.

Meanwhile, in the positional relationships along the ejection surface (XY plane), the damper mechanisms **301** overlap neither the supply connection flow paths **15a** nor the discharge connection flow paths **15b** on the XY plane. Note that the positional relationship between the damper mechanisms **301** and each of the supply connection flow paths **15a** and the discharge connection flow paths **15b** is not limited to that in the example in FIG. **4**.

(33) A configuration of the liquid ejection substrate **2** illustrated in FIGS. **3A** and **3B** and FIG. **4** is an example of a liquid ejection substrate to which the present invention is applicable, and a configuration of the liquid ejection substrate to which the present invention is applicable is not limited to that in this example. For example, in the liquid ejection substrate **2** illustrated in FIGS. **3A** and **3B** and FIG. **4**, the ink supplied from the external ink tank to the liquid ejection substrate **2** passes through the pressure chambers **11**, and a part of the ink is ejected from the nozzles **3**, while the other part thereof circulates so as to return to the external ink tank. However, the present invention is also applicable to the liquid ejection substrate **2** not having such a circulation flow path. Each of the following first and second embodiments is an example in which the present invention is applied to the liquid ejection substrate **2** having no circulation flow path, while each of the following third and fourth embodiments is an example in which the present invention is applied to the liquid ejection substrate **2** having a circulation flow path.

(34) In the liquid ejection substrate **2** illustrated in FIGS. **3A** and **3B** and FIG. **4**, in the direction (Z-direction) crossing the ejection surface, the common flow paths **13a** and **13b** and the damper mechanisms **301** are disposed opposite to the nozzles **3** with respect to the pressure chambers **11**. However, the present invention is also applicable to the liquid ejection substrate **2** in which, in the direction (Z-direction) crossing the ejection surface, the common flow path and the damper mechanisms are disposed on the same side as that of the nozzles with respect to the pressure chambers. Each of the following first, third, and fourth embodiments is the example in which the present invention is applied to the liquid ejection substrate **2** in which, in the direction (Z-direction) crossing the ejection surface, the common flow path and the damper mechanism are disposed opposite to the nozzles with respect to the pressure chambers. The second embodiment is the example in which the present invention is applied to the liquid ejection substrate **2** in which, in the direction (Z-direction) crossing the ejection surface, the common flow path and the damper mechanisms are disposed on the same side as that of the nozzles with respect to the pressure chambers.

First Embodiment

(35) Referring to FIGS. 5A and 5B, a description will be given of the liquid ejection substrate in the first embodiment of the present invention. FIG. 5A is a plan view illustrating a portion of the liquid ejection substrate **2** in the first embodiment. FIG. 5B is a cross-sectional view along an A-A line in FIG. 5A.

(36) In the liquid ejection substrate **2**, the plurality of nozzles **3** are arranged in the X-direction (first direction) to form one nozzle row, and the plurality of nozzle rows are arranged in the Y-direction (second direction). In FIG. 5A, a part of each of the two nozzle rows is illustrated.

(37) In the liquid ejection substrate **2**, the plurality of pressure chambers **11** are arranged along the X-direction. Sides of the pressure chambers **11** extending along the X-direction of the pressure chambers **11** are short sides, while sides thereof extending along the Y-direction are long sides. One end portions of the pressure chambers **11** in the longitudinal direction (Y-direction) are formed with the nozzles **3**, while other end portions thereof communicate with discrete flow paths **12**. The plurality of individual pressure chambers **11** communicate with the common flow path **13** via the discrete flow paths **12**. The plurality of discrete flow paths **12** communicating with the plurality of individual pressure chambers **11** included in the two nozzle rows illustrated in FIG. 5A communicate with the same common flow path **13**. The common flow path **13** communicates with connection flow paths **15**, and the connection flow paths **15** communicate with the external ink tank via external opening portions **20**.

(38) The common flow path **13** communicates with the discrete flow paths **12** via the discrete opening portions **120**, and each of the damper mechanisms **301** is disposed so as to face the discrete opening portions **120** in the direction (Z-direction) crossing the ejection surface.

(39) The common flow path **13** is formed with partition walls **16** disposed between the discrete opening portions **120** adjacent in the X-direction (direction in which the nozzles **3** are arranged or first direction) to extend in the direction (Z-direction) crossing the ejection surface. In the first embodiment, the partition walls **16** are not disposed between all the adjacent discrete opening portions **120**.

(40) The partition walls **16** extend in the Y-direction (second direction), while bending, and have wall surfaces crossing the X-direction (first direction). Thus, a pressure fluctuation propagating in the X-direction in the common flow path **13** is reflected by the partition walls **16**, and the pressure fluctuation is inhibited from propagating over the entire region of the common flow path **13** extending in the X-direction.

(41) One end portion of each of the partition walls **16** in the Y-direction is in contact with one side wall of the common flow path **13** extending along the X-direction (first direction), while another end portion thereof is apart from another side wall of the common flow path **13** extending along the X-direction (first direction). In the positional relationships along the ejection surface (XY plane), the partition walls **16** do not overlap the connection flow paths **15**. This can inhibit the partition walls **16** from excessively controlling a speed of a liquid flow in the common flow path **13**.

(42) In the Y-direction (second direction) crossing the X-direction (first direction) along the ejection surface, a length  $W2$  of each of the partition walls **16** is larger than a length  $W1$  of each of the discrete opening portions **120** ( $W2 > W1$ ). This can more reliably inhibit the pressure fluctuation from propagating between the adjacent discrete opening portions **120**.

(43) In FIG. 5B, the liquid ejection substrate **2** is formed to have a multilayer configuration including the nozzle substrate **201**, the vibration substrate **202**, the liquid supply substrate **203**, and the flow path formation substrate **204**. Each of the substrates may be configured to include a single layer or may also be configured to have a multilayer structure including a plurality of layers or a plurality of substrates.

(44) The nozzle substrate **201** is formed with the nozzles **3** and the pressure chambers **11**.

(45) The vibration substrate **202** is formed with depressed portions **24**, and the vibration substrate **202** is fixed to the nozzle substrate **201** via vibration plates **17**. In spaces formed by the vibration plates **17** and the depressed portions **24**, piezoelectric elements **18** serving as actuators that deform



the vibration plates **17** are provided. The vibration substrate **202** is provided with the discrete flow paths **12** and, at positions corresponding to positions of the discrete flow paths **12**, through holes **121** are formed in the vibration plates **17** to provide communication between the pressure chambers **11** and the discrete flow paths **12**.

(46) The liquid supply substrate **203** is provided with the common flow path **13** and the partition walls **16**, and the liquid supply substrate **203** is fixed to the vibration substrate **202** to provide communication between the discrete flow paths **12** and the common flow path **13**.

(47) The flow path formation substrate **204** is formed with the connection flow paths **15** extending along the first direction (X-direction) and the damper mechanism **301**, and the flow path formation substrate **204** is fixed to the liquid supply substrate **203** via an adhesion layer **19**. The adhesion layer **19** is not formed in a region corresponding to the common flow path **13**, the liquid in the common flow path **13** is in contact with the damper member **300**, and the common flow path **13** and the connection flow paths **15** communicate with each other. The damper member **300** is a flexible member that is flexibly deformed with the pressure fluctuation in the liquid in the common flow path **13**. A space **22** serving as the damper chamber and the damper member **300** provided so as to have one surface in contact with the liquid in the common flow path **13** and another surface in contact with a gas in the space **22** are included in each of the damper mechanisms **301** that absorb the pressure fluctuation in the liquid in the common flow path **13**. The common flow path **13** and the damper mechanisms **301** are formed of the liquid supply substrate **203** serving as a common flow path substrate that forms the side walls of the common flow path **13** and the partition walls **16** and the flow path formation substrate **204** serving as a damper substrate including the damper mechanisms **301** which are fixed to each other via the adhesion layer **19**. In a region where the common flow path **13** and the connection flow paths **15** are connected, a through hole or a slit may also be provided in the damper member **300** to form a filter **21**. Between each of portions of the liquid supply substrate **203** serving as the common flow path substrate that form the partition walls **16** and the flow path formation substrate **204** serving as the damper substrate, the adhesion layer **19** is not formed, and the partition walls **16** are thereby configured so as not to come into contact with the damper members **300**. This can inhibit the presence of the partition walls **16** from inhibiting the flexible deformation or vibration of the damper members **300**.

(48) In the liquid ejection substrate **2** in the first embodiment, when a voltage is applied to any of the piezoelectric elements **18** from electric wiring (not shown), the vibration plate **17** is deformed to cause a pressure fluctuation in the liquid in the pressure chamber **11**. By applying, to the piezoelectric element **18**, the voltage according to a resonance frequency of the pressure chamber **11** containing a liquid **14** and displacing the vibration plate **17** by combining together a direction in which the space in the pressure chamber **11** is enlarged and a direction in which the space in the pressure chamber **11** is reduced, droplets of the liquid **14** are ejected from the nozzles **3**.

Accordingly, it is possible to eject the liquid **14** from the nozzles **3** in response to a drive signal input to the piezoelectric element **18**. The liquid **14** is supplied from the liquid tank (not shown) to the pressure chamber **11** via the external opening portion **20**, the connection flow path **15**, the common flow path **13**, and the discrete flow path **12**.

(49) The displacement of the vibration plate **17** resulting from the application of the voltage to the piezoelectric element **18** and the ejection of the liquid **14** from the nozzles **3** causes the pressure fluctuation in the liquid **14** in the pressure chamber **11**. This pressure fluctuation propagates to another of the pressure chambers **11** communicating via the discrete flow path **12** and the common flow path **13**. When the liquid **14** is ejected in a state where the pressure fluctuation has occurred, an intended change may occur in characteristics of ejected droplets such as a speed of the ejected liquid **14**, a volume thereof, and a state of coherence of droplets thereof.

(50) In this respect, in the liquid ejection substrate **2** in the first embodiment, each of the damper mechanisms **301** is disposed at a position facing the discrete opening portion **120** in the common flow path **13** in the direction (Z-direction) crossing the ejection surface, and the pressure

fluctuation that has propagated to the common flow path **13** can be reduced using the damper mechanism **301**. In addition, between some of the adjacent discrete opening portions **120**, the partition walls **16** are provided. Thus, the pressure fluctuation propagating in the X-direction (first direction or longitudinal direction of the common flow path **13**) in the space of the common flow path **13** is cut off at positions of the partition walls **16**, and the pressure fluctuation is inhibited from propagating over a long distance in the common flow path **13**. Since the partition walls **16** are not in contact with the damper member **300**, it is possible to inhibit the presence of the partition walls **16** from affecting an effect of reducing the pressure fluctuation achieved by the damper member **300**. Thus, the synergetic effect of the partition walls **16** that cut off the pressure fluctuation propagating in the common flow path **13** and the damper mechanisms **301** that absorb the pressure fluctuation can inhibit the pressure fluctuation occurring in a given one of the pressure chambers **11** from affecting another of the pressure chambers **11**. As a result, it is possible to reduce crosstalk between the pressure chambers **11** communicating with each other via the common flow path **13** and inhibit unintended fluctuations of the characteristics of the droplets of the liquid **14** ejected from the nozzles **3**. This can inhibit quality degradation of printing performed by the recording apparatus **101** having the liquid ejection head module **1** on the recording medium **111**.

(51) Note that, in the first embodiment, as each of the damper mechanisms **301**, a configuration including the damper member **300** and the space **22** provided in the flow path formation substrate **204** is illustrated by way of example. However, the damper mechanism **301** is not limited to this configuration as long as the damper mechanism **301** has a configuration that reduces the pressure fluctuation by using the deformation of the damper member **300**.

#### Second Embodiment

(52) Referring to FIGS. **6A** and **6B**, a description will be given of the liquid ejection substrate in the second embodiment of the present invention. Note that the following description will be given mainly of differences from the first embodiment, and a detailed description of the same configuration as that in the first embodiment is omitted by using the same reference signs as those used in the first embodiment.

(53) FIG. **6A** is a plan view illustrating a part of the liquid ejection substrate **2** in the second embodiment. FIG. **6B** is a cross-sectional view along the line A-A in FIG. **6A**. FIG. **6A** illustrates a part of each of two nozzle rows **N1** and **N2** in the same manner as in FIG. **5A**.

(54) The discrete flow paths **12** communicating with the pressure chambers **11** in which the nozzles **3** in the adjacent nozzle rows **N1** and **N2** are formed communicate with the same common flow path **13**. In an end portion of the common flow path **13** in the X-direction, an external opening portion **20** is formed to be connected to a liquid tank (not shown).

(55) The common flow path **13** communicates with the discrete flow paths **12** via the discrete opening portions **120**, and the damper mechanisms **301** are disposed so as to face the discrete opening portions **120** in the direction (Z-direction) crossing the ejection surface.

(56) It is assumed that a row of the discrete opening portions **120** corresponding to the nozzles **3** in the first nozzle row **N1** is a first discrete opening portion row **R1**, while a row of the discrete opening portions **120** corresponding to the nozzles **3** in the second nozzle row **N2** is a second discrete opening portion row **R2**.

(57) The partition walls **16** include partition walls **161** disposed between the discrete opening portions **120** in the first discrete opening portion row **R1** which are adjacent in the X-direction (direction in which the nozzles **3** are arranged or first direction) and partition walls **162** disposed between the discrete opening portions **120** in the second discrete opening portion row **R2** which are adjacent in the X-direction. Each of the partition walls **161** extends in the Y-direction to a position not overlapping the discrete opening portion **120** in the second discrete opening portion row **R2**, while each of the partition walls **162** extends in the Y-direction to a position not overlapping the discrete opening portion **120** in the first discrete opening portion row **R1**. The partition walls **161** and the partition walls **162** have respective positions in the Y-direction which partly overlap each

other, and are staggered in the positional relationships along the ejection surface (XY plane) to be arranged in the form of comb teeth as a whole.

(58) The partition walls **161** and **162** extend in the Y-direction (second direction) to have wall surfaces crossing the X-direction (first direction). Thus, the pressure fluctuation propagating in the X-direction in the common flow path **13** is reflected by the partition walls **161** and **162**, and the pressure fluctuation is inhibited from propagating over the entire region of the common flow path **13** extending in the X-direction.

(59) The partition walls **161** and **162** are apart from wall surfaces of the common flow path **13** extending along the X-direction. This can inhibit the partition walls **161** and **162** from excessively controlling the speed of the liquid flow in the common flow path **13**.

(60) In the Y-direction (second direction), lengths of the partition walls **161** and **162** are larger than lengths of the discrete opening portions **120**. This can more reliably inhibit the pressure fluctuation from propagating between the adjacent discrete opening portions **120**.

(61) In FIG. **6B**, the liquid ejection substrate **2** is formed to have the multilayer configuration including the nozzle substrate **201**, the flow path formation substrate **204**, the liquid supply substrate **203**, and the vibration substrate **202**. Each of the substrates may be configured as a single layer or may also be configured to have a multilayer structure including a plurality of layers or a plurality of substrates.

(62) The nozzle substrate **201** is formed with the nozzles **3** and the space **22**.

(63) The flow path formation substrate **204** is formed with the common flow path **13** and the partition walls **16**, and the flow path formation substrate **204** is fixed to the nozzle substrate **201** via the adhesion layer **19**. The adhesion layer **19** is not formed in the region corresponding to the common flow path **13**, and the liquid in the common flow path **13** is in contact with the damper member **300**. The damper member **300** is the flexible member that is flexibly deformed with the pressure fluctuation in the liquid in the common flow path **13**. The space **22** and the damper member **300** provided so as to have the one surface in contact with the liquid in the common flow path **13** and the other surface in contact with the gas in the space **22** are included in each of the damper mechanisms **301** that absorb the pressure fluctuation in the liquid in the common flow path **13**. The common flow path **13** and the damper mechanisms **301** are formed of the flow path formation substrate **204** serving as the common flow path substrate that forms the side walls of the common flow path **13** and the partition walls **16** and the nozzle substrate **201** serving as the damper substrate including the damper mechanisms **301** which are fixed to each other via the adhesion layer **19**.

(64) The liquid supply substrate **203** is formed with the discrete flow paths **12** and the pressure chambers **11**, and the liquid supply substrate **203** is fixed to the flow path formation substrate **204** to provide communication between the discrete flow paths **12**, the common flow path **13**, and the pressure chambers **11**.

(65) In the vibration substrate **202**, the depressed portions **24** are formed, and the vibration substrate **202** is fixed to the liquid supply substrate **203** via the vibration plates **17**. In the spaces formed by the vibration plates **17** and the depressed portions **24**, the piezoelectric elements **18** serving as the actuators that deform the vibration plates **17** are provided.

(66) Through flow paths **23** providing communication between the nozzles **3** and the pressure chambers **11** are formed in the nozzle substrate **201**, the flow path formation substrate **204**, and the liquid supply substrate **203**.

(67) In the second embodiment, the liquid **14** from the liquid tank (not shown) is ejected from the nozzles **3** via the external opening portions **20**, the common flow path **13**, the discrete flow paths **12**, the pressure chambers **11**, and the through flow paths **23**. The liquid **14** supplied to the common flow path **13** is supplied to the pressure chambers **11** through the spaces between the partition walls **161** and **162** arranged in the form of comb teeth.

(68) In the second embodiment, the partition walls **16** are disposed between the discrete opening

portions **120** adjacent in the X-direction, and therefore it is possible to inhibit the propagation of the pressure fluctuation to the pressure chamber **11** adjacent in the X-direction, which is significantly affected by the propagation of the pressure fluctuation. In addition, since the partition walls **16** are arranged in the form of comb teeth in a middle portion of the common flow path **13** in the Y-direction, and are not provided in the vicinities of the both end portions of the common flow path in the Y-direction, and open spaces are provided, the pressure fluctuation propagates in the vicinities of the both end portions of the common flow path in the Y-direction. Accordingly, it is also possible to inhibit the propagation of the pressure fluctuation to the pressure chamber **11** adjacent in the Y-direction.

(69) In the second embodiment also, between each of the partition walls **16** and the damper member **300**, the adhesion layer **19** is not formed, and the partition wall **16** and the damper member **300** are not in contact with each other, and therefore it is possible to inhibit the presence of the partition walls **16** from affecting the effect of reducing the pressure fluctuation achieved by the damper member **300**. The synergetic effect of the partition walls **16** that cut off the pressure fluctuation propagating in the common flow path **13** and the damper mechanisms **301** that absorb the pressure fluctuation can inhibit the pressure fluctuation occurring in a given one of the pressure chambers **11** from affecting another of the pressure chambers **11**. As a result, it is possible to reduce the crosstalk between the pressure chambers **11** communicating with each other via the common flow path **13** and inhibit the unintended fluctuations of the characteristics of the droplets of the liquid **14** ejected from the nozzles **3**. This can inhibit the quality degradation of the printing performed by the recording apparatus **101** having the liquid ejection head module **1** on the recording medium **111**.

(70) Note that, in the second embodiment, as each of the damper mechanisms **301**, a configuration including the damper member **300** and the space **22** provided in the nozzle substrate **201** is illustrated by way of example. However, the damper mechanism **301** is not limited to this configuration as long as the damper mechanism **301** has a configuration that reduces the pressure fluctuation by using the deformation of the damper member **300**. For example, it may also be possible to adopt a configuration in which an opening is provided in a surface of the nozzle substrate **201** facing the damper member **300** via the space **22** or a configuration in which the damper member **300** is not provided, but a nozzle-shaped opening is provided in a wall surface between the space **22** and the common flow path **13** to form a meniscus for the liquid **14**.

### Third Embodiment

(71) Referring to FIGS. 7A and 7B, a description will be given of the liquid ejection substrate in the third embodiment of the present invention. Note that the following description will be given mainly of differences from the first embodiment, and a detailed description of the same configuration as that in the first embodiment is omitted by using the same reference signs as those used in the first embodiment.

(72) FIG. 7A is a plan view illustrating a part of the liquid ejection substrate **2** in the third embodiment. FIG. 7B is a cross-sectional view along the line A-A in FIG. 7A. FIGS. 7A and 7B illustrate a part of each of the four nozzle rows **N1**, **N2**, **N3**, and **N4**.

(73) In the vicinity of the middle of each of the pressure chambers **11** in the Y-direction (longitudinal direction of the pressure chamber **11**), the nozzles **3** are formed to communicate with each of the discrete supply flow path **12a** and the discrete discharge flow path **12b** in the vicinities of the both end portions thereof in the Y-direction. The pressure chambers **11** communicate with the common supply flow paths **13a** and the supply connection flow paths **15a** via the discrete supply flow paths **12a**, while the supply connection flow paths **15a** communicate with an external ink tank via the external supply opening portions **20a**. The pressure chambers **11** are connected to the common discharge flow paths **13b** and the discharge connection flow paths **15b** via the discrete discharge flow paths **12b**, while the discharge connection flow paths **15b** communicate with the external ink tank via the external discharge opening portions **20b**.

(74) The common supply flow paths **13a** communicate with the discrete supply flow paths **12a** via the discrete supply opening portions **120a**, and the damper mechanisms **301** are disposed so as to face the discrete supply opening portions **120a** in the direction (Z-direction) crossing the ejection surface. The common discharge flow paths **13b** communicate with the discrete discharge flow paths **12b** via the discrete discharge opening portions **120b**, and the damper mechanisms **301** are disposed so as to face the discrete discharge opening portions **120b** in the direction (Z-direction) crossing the ejection surface.

(75) In the third embodiment, the liquid **14** supplied from the liquid tank (not shown) circulates in the liquid tank via the external supply opening portions **20a**, the supply connection flow paths **15a**, the pressure chambers **11**, the discharge connection flow paths **15b**, and the external discharge opening portions **20b**. Such circulation of the liquid **14** is implemented by, e.g., providing a predetermined pressure difference between the supply connection flow paths **15a** and the discharge connection flow paths **15b**. By causing the liquid **14** to circulate, it is possible to inhibit an increase in the vicinity of the liquid **14** in the vicinities of the nozzles **3** due to vaporization of the liquid **14** from the nozzles **3**. In a configuration in the third embodiment, structures of a supply system and a discharge system are symmetrical, and accordingly it is also possible to reverse a direction of the circulation of the liquid **14**.

(76) The discrete supply flow paths **12a** communicating with the pressure chambers **11** in which the nozzles **3** in the adjacent nozzles rows **N1** and **N2** are formed communicate with the same common supply flow path **13a**. The discrete discharge flow paths **12b** communicating with the pressure chambers **11** in which the nozzles **3** in the adjacent nozzle rows **N2** and **N3** are formed communicate with the same common discharge flow path **13b**. The same applies also to the adjacent nozzle rows **N3** and **N4**.

(77) The common supply flow paths **13a** communicate with the discrete supply flow paths **12a** via the discrete supply opening portions **120a**, and the damper mechanisms **301** are disposed so as to face the discrete supply opening portions **120a** in the direction (Z-direction) crossing the ejection surface.

(78) The common discharge flow paths **13b** communicate with the discrete discharge flow paths **12b** via the discrete discharge opening portions **120b**, and the damper mechanisms **301** are disposed so as to face the discrete discharge opening portions **120** in the direction (Z-direction) crossing the ejection surface.

(79) In the third embodiment, in the direction (Z-direction) crossing the ejection surface, the common flow paths **13a** and **13b** and the damper mechanisms **301** are disposed opposite to the nozzles **3** with respect to the pressure chambers **11**. In other words, a configuration is provided in which the common flow paths **13a** and **13b** and the damper mechanisms **301** are provided in the liquid supply substrate **203** located opposite to the nozzle substrate **201** with respect to the vibration substrate **202**. The flow paths (which are the through flow paths **23** in the second embodiment) connecting the nozzles **3** and the pressure chambers **11** are shorter than those in the configuration in the second embodiment (FIGS. **6A** and **6B**) in which the common flow path **13** and the damper mechanism **301** are provided on the same side as that of the nozzle substrate **201** with respect to the vibration substrate **202**. Consequently, it is possible to more efficiently guide a circulating flow of the liquid **14** to the nozzles **3**. Therefore, it is possible to more effectively inhibit the increased viscosity of the liquid **14** in the nozzles **3**.

(80) In the common supply flow paths **13a**, the partition walls **16** are formed to be disposed between the discrete supply opening portions **120a** adjacent in the X-direction (direction in which the nozzles **3** are arranged or first direction) and extend in the direction (Z-direction) crossing the ejection surface. Partition walls **16N1** disposed between the discrete supply opening portions **120a** corresponding to the nozzle row **N1** and partition walls **16N2** disposed between the discrete supply opening portions **120a** corresponding to the nozzle row **N2** at a position adjacent to the discrete supply opening portions **120a** are formed integrally.

(81) Likewise, in the common discharge flow paths **13b**, the partition walls **16** are formed to be disposed between the discrete discharge opening portions **120b** adjacent in the X-direction (direction in which the nozzles **3** are arranged or first direction) and extend in the direction (Z-direction) crossing the ejection surface.

(82) The partition walls **16** extend in the Y-direction (second direction), while bending, and the wall surfaces thereof cross the X-direction (first direction). Thus, a pressure fluctuation propagating in the X-direction in the common supply flow paths **13a** and the common discharge flow paths **13b** is reflected by the partition walls **16**, and the pressure fluctuation is inhibited from propagating over the entire region of the common supply flow paths **13a** and the common discharge flow paths **13b** each extending in the X-direction.

(83) One end portion of each of the partition walls **16** in the Y-direction is in contact with one side wall of the common supply flow path **13a** extending along the X-direction (first direction), while another end portion thereof is apart from another wall surface of the common supply flow path **13a** extending along the X-direction (first direction). In the positional relationships along the ejection surface (XY plane), the partition walls **16** do not overlap the supply connection flow paths **15a**.

(84) One end portion of each of the partition walls **16** in the Y-direction is in contact with one side wall of the common discharge flow path **13b** extending along the X-direction (first direction), while another end portion thereof is apart from another wall surface of the common discharge flow path **13b** extending along the X-direction (first direction). In the positional relationships along the ejection surface (XY plane), the partition walls **16** do not overlap the discharge connection flow paths **15b**.

(85) This can inhibit the partition walls **16** from excessively controlling a speed of a liquid flow in the common supply flow paths **13a** and the common discharge flow paths **13b**.

(86) In FIG. 7B, the liquid ejection substrate **2** is formed to have the multilayer configuration including the nozzle substrate **201**, the vibration substrate **202**, the liquid supply substrate **203**, and the flow path formation substrate **204**. Each of the substrates may be configured as a single layer or may also be configured to have a multilayer structure including a plurality of layers or a plurality of substrates.

(87) In the third embodiment, the pressure fluctuation occurring in the pressure chamber **11** into which the liquid **14** is ejected propagates to the common supply flow path **13a** via the discrete supply flow path **12a**, and propagates to the common discharge flow path **13b** via the discrete discharge flow path **12b**. However, the partition walls **16** cut off the pressure fluctuation propagating in the X-direction (first direction or direction in which the nozzles **3** are arranged) in the common supply flow path **13a** and the common discharge flow path **13b**. Thus, the pressure fluctuation is inhibited from propagating over a long distance in the common supply flow path **13a** and the common discharge flow path **13b**. In addition, the damper mechanism **301** absorbs the pressure fluctuation, and consequently influence of the propagation of the pressure fluctuation in the Y-direction (second direction or direction in which the nozzle rows N1, N2, . . . are arranged) is also reduced. In the same manner as in the other embodiments, each of the partition walls **16** and the damper member **300** are not in contact with each other, and therefore it is possible to inhibit the presence of the partition walls **16** from affecting the effect of reducing the pressure fluctuation achieved by the damper member **300**. Thus, the synergetic effect of the partition walls **16** that cut off the pressure fluctuation propagating in the common flow path **13** and the damper mechanisms **301** that absorb the pressure fluctuation can inhibit the pressure fluctuation occurring in a given one of the pressure chambers **11** from affecting another of the pressure chambers **11**. As a result, it is possible to reduce the crosstalk between the pressure chambers **11** communicating with each other via the common flow path **13** and inhibit the unintended fluctuations of the characteristics of the droplets of the liquid **14** ejected from the nozzles **3**. This can inhibit quality degradation of the printing performed by the recording apparatus **101** having the liquid ejection head module **1** on the recording medium **111**.

(88) Note that, by controlling the timing of the ejection of the liquid **14** such that the liquid **14** is ejected at different timings in the adjacent nozzle rows, it is also possible to further reduce the influence of the crosstalk between the adjacent nozzle rows.

#### Fourth Embodiment

(89) Referring to FIGS. **8A** and **8B**, a description will be given of the liquid ejection substrate in the fourth embodiment of the present invention. Note that the following description will be given mainly of differences from the first embodiment, and a detailed description of the same configuration as that in the first embodiment is omitted by using the same reference signs as those used in the first embodiment.

(90) FIG. **8A** is a plan view illustrating a part of the liquid ejection substrate **2** in the fourth embodiment. FIG. **8B** is a cross-sectional view along the line A-A in FIG. **8A**. FIGS. **8A** and **8B** illustrate a part of each of the four nozzle rows **N1**, **N2**, **N3**, and **N4**.

(91) In contrast to the third embodiment, in the fourth embodiment, the damper mechanisms **301** are not disposed in the common supply flow paths **13a**, but are disposed only in the common discharge flow paths **13b**. A dimension of each of the damper mechanisms **301** in the Y-direction (direction in which the nozzle rows are arranged) is larger than that in the third embodiment. Accordingly, it is possible to obtain a larger amplitude of the damper member **300**. The common discharge flow paths **13b** communicate with the discrete discharge flow paths **12b** via the discrete discharge opening portions **120b**, and the damper mechanisms **301** are disposed so as to face the discrete discharge opening portions **120b** in the direction (Z-direction) crossing the ejection surface.

(92) The discrete supply flow paths **12a** and the discrete discharge flow paths **12b** are designed such that respective flow path resistances thereof are equal, and a pressure difference is provided such that a pressure in each of the discharge connection flow paths **15b** is lower than that in each of the supply connection flow paths **15a**. As a result, a circulating flow of the liquid **14** flowing from the supply connection flow path **15a** to the discharge connection flow path **15b** via the discrete supply flow path **12a**, the pressure chamber **11**, and the discrete discharge flow path **12b** is formed. In this case, the pressure fluctuation caused by the ejection is more likely to propagate to the discrete discharge flow path **12b** at a relatively low pressure than to the discrete supply flow path **12a** at a relatively high pressure. Therefore, in a configuration in which the damper mechanisms **301** are provided in either of the common supply flow paths **13a** and the common discharge flow paths **13b**, the damper mechanisms **301** disposed in the common discharge flow paths **13b** allow a higher crosstalk inhibiting effect to be obtained. When the damper mechanisms **301** are disposed in the common discharge flow paths **13b**, it may also be possible to design each of the discrete supply flow paths **12a** and the discrete discharge flow paths **12b** such that the flow path resistance of the discrete discharge flow path **12b** is smaller than that of the discrete supply flow path **12a** and that the pressure fluctuation due to the ejection is more likely to propagate to the common discharge flow paths **13b**.

(93) In the common supply flow paths **13a**, the plurality of external supply opening portions **20a** to which the liquid is supplied from the outside are arranged along the X-direction. In the common discharge flow paths **13b**, the plurality of external discharge opening portions **20b** through which the liquid is discharged to the outside are arranged along the X-direction.

(94) In the fourth embodiment, in the common flow paths in which the damper mechanisms **301** are disposed, the partition walls are provided. In the fourth embodiment, the damper mechanisms **301** are disposed in the common discharge flow paths **13b**, and accordingly the partition walls **16** are provided in the common discharge flow paths **13b**. Specifically, in the common discharge flow paths **13b**, the partition walls **16** are formed to be disposed between the discrete discharge opening portions **120b** adjacent in the X-direction (direction in which the nozzles **3** are arranged or first direction) and extend in the direction (Z-direction) crossing the ejection surface. In the fourth embodiment, in the same manner as in the first embodiment, the partition walls **16** are not disposed

between all the adjacent discrete discharge opening portions **120b**, but are formed at positions of the external discharge opening portions **20b** in the X direction and at positions between the external discharge opening portions **20b** in the X direction. Accordingly, intervals between the plurality of partition walls **16** in the X-direction (first direction) are smaller than intervals between the plurality of external discharge opening portions **20b** in the X-direction.

(95) Note that it may also be possible to adopt a configuration in which the damper mechanisms **301** are disposed only in the common supply flow paths **13a**. In that case, it may also be possible to provide the partition walls **16** in the common supply flow paths **13a** and form the partition walls **16** at positions of the external supply opening portions **20a** in the X-direction and at positions between the external supply opening portions **20a** in the X-direction. In this case, the intervals between the plurality of partition walls **16** in the X-direction (first direction) are smaller than the intervals between the plurality of external supply opening portions **20a** in the X-direction. In the first embodiment also, when the plurality of external opening portions **20** are provided along the X-direction (first direction) in the connection flow paths **15**, the positional relationship between the external opening portions **20** and the partition walls **16** in the X-direction may also be the same as that in the fourth embodiment. In other words, the intervals between the plurality of partition walls **16** in the X-direction (first direction) may also be set smaller than the intervals between the plurality of external opening portions **20** in the X-direction. The positions of the partition walls **16** in the X-direction may also be positions of the external opening portions **20** in the X-direction and positions between the external opening portions **20** in the X-direction.

(96) The partition walls **16** extend in the Y-direction (second direction), while bending, and have the wall surfaces crossing the X-direction (first direction). Thus, at the positions at which the partition walls **16** are provided, spaces in the common discharge flow paths **13b** are partially partitioned in the X-direction. The pressure fluctuation propagating in the X-direction in the common discharge flow paths **13b** is reflected by the partition walls **16**, and the pressure fluctuation is inhibited from propagating over the entire region of the common flow path **13** extending in the X-direction. Note that, in the same manner as in the first embodiment, in each of portions in which the partition walls **16** are formed, the adhesion layer **19** is not formed such that vibration absorption performance of the damper member **300** is not inhibited by the partition walls **16**.

(97) One end portion of each of the partition walls **16** in the Y-direction is in contact with one side wall of the common discharge flow path **13b** extending along the X-direction (first direction), while another end portion thereof is apart from another side wall of the common discharge flow path **13b** extending along the X-direction (first direction). In the positional relationships along the ejection surface (XY plane), the partition walls **16** do not overlap the discharge connection flow paths **15b**. This can inhibit the partition walls **16** from excessively controlling a speed of a liquid flow in the common discharge flow path **13b**.

(98) In the fourth embodiment, in the same manner as in the first embodiment, the liquid supply substrate **203** is formed with the common supply flow paths **13a**, the common discharge flow paths **13b**, and the partition walls **16**. The flow path formation substrate **204** is formed with the supply connection flow paths **15a**, the discharge connection flow paths **15b**, and the damper mechanisms **301** each extending along the first direction (X-direction). The flow path formation substrate **204** is fixed to the liquid supply substrate **203** via the adhesion layer **19**. The adhesion layer **19** is not formed in regions corresponding to the common supply flow paths **13a** and the common discharge flow paths **13b**, and the liquid in the common discharge flow paths **13b** is in contact with the damper member **300**. Each of the common flow paths **13a** and **13b** and the damper mechanisms **301** are formed of the liquid supply substrate **203** serving as the common flow path substrate that forms the side walls of the common flow path and the partition walls **16** and the flow path formation substrate **204** serving as the damper substrate including the damper mechanisms **301** which are fixed to each other via the adhesion layer **19**. Between the portions of the liquid supply



substrate **203** serving as the common flow path substrate that form the partition walls **16** and the flow path formation substrate **204** serving as the damper substrate, the adhesion layer **19** is not formed, and accordingly the partition walls **16** are configured so as not to come into contact with the damper member **300**. In addition, between the portions of the liquid supply substrate **203** serving as the common flow path substrate that form side walls **130** separating the common supply flow paths **13a** and the common discharge flow paths **13b** from each other and the flow path formation substrate **204** serving as the damper substrate also, the adhesion layer **19** is not formed. As a result, the common supply flow paths **13a** and the common discharge flow paths **13b** communicate with each other.

(99) An advantage of not forming the adhesion layer **19** in each of the portions in which the side walls **130** separating the common supply flow paths **13a** and the common discharge flow paths **13b** from each other are formed is that a width of each of the damper mechanisms **301** in the Y-direction can be ensured. In other words, to form the adhesion layer **19** in a portion having a small width in the Y-direction, such as each of the side walls **130** separating the common supply flow paths **13a** and the common discharge flow paths **13b** from each other, according to accuracy of formation of the adhesion layer **19**, the width of the side wall **130** in the Y-direction needs to have a given or larger dimension. To ensure the width of the side wall **130** in the Y-direction, it is necessary to accordingly reduce a width of each of the damper mechanisms **301** in the Y-direction, resulting in degradation of the vibration absorption performance of the damper mechanism **301**. By not forming the adhesion layer **19** in each of the portions with the side walls **130**, it is possible to ensure the width of the damper mechanism **301** in the Y-direction and ensure the vibration absorption performance.

(100) When the adhesion layer **19** is not formed in each of the portions with the side walls **130**, in the portion with the side wall **130**, a communication path **131** corresponding to a thickness of the adhesion layer **19** is formed between the common supply flow path **13a** and the common discharge flow path **13b**. Since the communication path **131** has a sufficiently high flow path resistance, by providing a predetermined pressure difference between the supply connection flow path **15a** and the discharge connection flow path **15b**, even when the communication path **131** is present, it is possible to obtain a sufficient circulating flow speed of the liquid **14**. Therefore, it is possible to inhibit the increased viscosity of the liquid **14** in the nozzles **3** due to the vaporization of the liquid **14**.

(101) Note that, in a portion with a side wall **132** having a large width in the Y-direction among the side walls separating the common supply flow paths **13a** and the common discharge flow paths **13b** from each other, the adhesion layer **19** may be formed in the same manner as in the other embodiments.

(102) In the fourth embodiment, the damper member **300** is divided in the X-direction into a plurality of portions **301a**, **301b**, and **301c**. This is because the width (dimension in the Y-direction) of each of the damper mechanisms **301** is sufficiently large and, even when the damper member **300** is divided in the X-direction, a sufficient vibration absorbing effect can be obtained. By dividing the damper member **300** in the longitudinal direction (X-direction) of the space in each of the common discharge flow paths **13b**, it is possible to inhibit the damper member **300** that is elongated in the X-direction from excessively vibrating. This can reduce the crosstalk propagating in the longitudinal direction (X-direction) of the space in the common discharge flow path **13b**.

(103) The foregoing configuration can inhibit a pressure fluctuation in the pressure chamber **11** caused by the ejection of the liquid **14** resulting from the driving of the piezoelectric element **18** by using a synergetic effect of the damper mechanism **301** and the partition wall **16** in each of the common discharge flow paths **13b**. Therefore, it is possible to inhibit the propagation of the pressure fluctuation to the communicating pressure chamber **11** and reduce fluctuations in the ejection characteristics of the liquid ejected from each of the nozzles **3**, which can reduce color unevenness of a printed image on the recording medium.

(104) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(105) This application claims the benefit of Japanese Patent Application No. 2022-48587, filed on Mar. 24, 2022, which is hereby incorporated by reference herein in its entirety.

## Claims

1. A liquid ejection head configured to eject a liquid from an ejection surface, the liquid ejection head comprising: a plurality of nozzles arranged along a first direction of the ejection surface; a plurality of pressure chambers communicating with the plurality of individual nozzles and provided with actuators configured to apply, to the liquid, pressures for ejecting the liquid from the nozzles; a plurality of discrete flow paths communicating with the plurality of individual pressure chambers; a common flow path communicating with the plurality of individual discrete flow paths via discrete opening portions and extending in the first direction; and a damper mechanism disposed to face the common flow path in a direction crossing the ejection surface to absorb a pressure fluctuation in the liquid in the common flow path, the common flow path being provided with a partition wall disposed between the discrete opening portions adjacent in the first direction to extend in the direction crossing the ejection surface.
2. The liquid ejection head according to claim 1, wherein a plurality of the partition walls are arranged along the first direction.
3. The liquid ejection head according to claim 1, wherein the partition wall is kept from contact with the damper mechanism.
4. The liquid ejection head according to claim 1, wherein, in a second direction extending along the ejection surface and crossing the first direction, a length of the partition wall is longer than a length of each of the discrete opening portions.
5. The liquid ejection head according to claim 1, wherein, in the direction crossing the ejection surface, each of the common flow path and the damper mechanism is disposed opposite to the nozzles with respect to the pressure chambers.
6. The liquid ejection head according to claim 1, wherein the common flow path and the damper mechanism are formed of a common flow path substrate and a damper substrate which are fixed to each other via an adhesion layer, the common flow path substrate forming each of a side wall of the common flow path and the partition wall, and the damper substrate including the damper mechanism, and wherein, between a portion of the common flow path substrate that forms the partition wall and the damper substrate, the adhesion layer is not provided.
7. The liquid ejection head according to claim 1, wherein the damper mechanism includes a damper chamber extending along the first direction and a flexible member provided so as to have one surface in contact with the liquid in the common flow path and another surface in contact with a gas in the damper chamber.
8. The liquid ejection head according to claim 1, wherein, in the direction crossing the ejection surface, each of the common flow path and the damper mechanism is disposed on the same side as that of the nozzles with respect to the pressure chambers.
9. The liquid ejection head according to claim 1, wherein the partition wall is disposed apart from a side wall of the common flow path extending along the first direction.
10. The liquid ejection head according to claim 1, wherein the common flow path is provided with a plurality of external opening portions arranged along the first direction to communicate with the outside, and wherein an interval in the first direction between a plurality of the partition walls arranged along the first direction is shorter than an interval in the first direction between the plurality of external opening portions.

11. The liquid ejection head according to claim 1, wherein the discrete flow paths include discrete supply flow paths configured to supply the liquid to the pressure chambers and discrete discharge flow paths configured to discharge the liquid from the pressure chambers, wherein the common flow path includes common supply flow paths that communicate with the plurality of discrete supply flow paths via discrete supply opening portions and common discharge flow paths that communicate with the plurality of discrete discharge flow paths via discrete discharge opening portions, wherein the damper mechanism is disposed so as to face at least either of the common supply flow path and the common discharge flow path in the direction crossing the ejection surface, and the partition wall is provided in the common flow path facing at least the damper mechanism.

12. The liquid ejection head according to claim 1, wherein each of the common supply flow paths, each of the common discharge flow paths, and the damper mechanism are formed of a common flow path substrate and a damper substrate which are fixed to each other via an adhesion layer, the common flow path substrate forming respective side walls of the common supply flow path and the common discharge flow path and the partition wall, and the damper substrate including the damper mechanism, and wherein, between a portion of the common flow path substrate that forms a side wall separating the common supply flow path and the common discharge flow path from each other and the damper substrate, the adhesion layer is not provided.

13. The liquid ejection head according to claim 1, wherein the common supply flow paths are provided with a plurality of external supply opening portions to which the liquid is supplied from the outside and which are arranged along the first direction, wherein the common discharge flow paths are provided with a plurality of external discharge opening portions which discharge the liquid to the outside and which are arranged along the first direction, and wherein an interval in the first direction between a plurality of the partition walls arranged along the first direction is shorter than an interval in the first direction between the plurality of external supply opening portions or smaller than an interval in the first direction between the plurality of external discharge opening portions.

14. A liquid ejection head configured to eject a liquid from an ejection surface, the liquid ejection head comprising: a plurality of nozzles arranged along a first direction of the ejection surface; a plurality of pressure chambers communicating with the plurality of individual nozzles and provided with actuators configured to apply, to the liquid, pressures for ejecting the liquid from the nozzles; a plurality of discrete flow paths communicating with the plurality of individual pressure chambers; a common flow path communicating with the plurality of individual discrete flow paths and extending in the first direction; and a damper mechanism disposed to face the common flow path in a direction crossing the ejection surface to absorb a pressure fluctuation in the liquid in the common flow path, the common flow path being provided with a partition wall that partially blocks a flow of the liquid along the first direction.

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